Md. Irtiza Hossain

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Education

Department of Computer Science and Engineering, BRAC University — Dhaka, Bangladesh
M.S. in Computer Science and Engineering (Specialization in Data Science) — Jun 2025 – Present

- Ongoing

Department of Computer Science and Engineering, BRAC University — Dhaka, Bangladesh
B.Sc. in Computer Science and Engineering — Dec 2019 – May 2024

- CGPA: 3.90 / 4.00 • Vice Chancellor's List (5 semesters), Dean's List (2 semesters)
- Thesis: *Enhanced CNN Approaches for Multi-Image Embedding in Image Steganography* (published at IEEE ICICT 2024)

Research Interests

3D Perception: LiDAR 3D object detection, scene understanding for autonomous systems.
Reliable AI: uncertainty estimation and calibration, human-in-the-loop decision support.
Efficient Deep Learning: model compression, mixture-of-experts, edge deployment.
Visual Reasoning: human behaviour analysis, graph neural networks, relational reasoning.

Teaching Experience

Lecturer, Department of Computer Science and Engineering — Nov 2024 – Present
BRAC University — Dhaka, Bangladesh

- Courses taught: Artificial Intelligence, Machine Learning, Data Structures, Algorithms, Computer Interfacing, Introduction to Robotics, Database Systems, and Systems Analysis & Design. Lectures, labs, and assessments.

Undergraduate Teaching Assistant — Oct 2022 – May 2024
BRAC University — Dhaka, Bangladesh

- Programming Language II (Object-Oriented Programming), Feb 2023 – May 2024; Differential Calculus and Coordinate Geometry, Oct – Dec 2022. Held office hours and graded examinations.

Industry Experience

Junior Software Engineer — Aug 2024 – Nov 2024
Orbit Informatics — Dhaka, Bangladesh

- Built and shipped a production Chrome extension (React, Node.js) for enterprise users.

Trainee Engineer (.NET) — Jun 2024 – Aug 2024
PRAN-RFL Group (othoba.com) — Dhaka, Bangladesh

- Automated abandoned-cart detection, adding 1–2% to recovered sales revenue.

Publications

Published

[1] **Edge-Native Digitization of Handwritten Marksheets: A Hybrid Heuristic-Deep Learning Framework**
Md. Irtiza Hossain, Junaid Ahmed Sifat, Abir Chowdhury, Md. Imran Bin Azad
IEEE Int. Conf. on Quantum Photonics, AI, and Networking (QPAIN), 2026. Presented.
Heuristic table parsing plus a pruned 2.1M-parameter YOLOv8 recognizer; 95.8% F1 and about 95× faster than PaddleOCR on a CPU. [IEEE Xplore] [Code]

[2] **Enhanced CNN Approaches for Multi-Image Embedding in Image Steganography**
Md. Irtiza Hossain, Samiul Kadir, Farhan Ishraq Fagun, Ishtiaq Samiul, Rafi Zaman Saukhin, Muhammad Iqbal Hossain
IEEE ICICT 2024. Presented.
Multi-scale CNN encoder–decoder that hides several secret images in one cover image, secured with AES-128 and ChaCha20. [IEEE Xplore] [Code]

Under Review and In Preparation

[3] **Predictive Self-Conditioning for Accurate and Calibrated LiDAR 3D Object Detection**
Sami Sadat, **Md. Irtiza Hossain**, Shreya Adrita Banik, Md Zia Uddin, Md. Golam Rabiul Alam
Under review, IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) 2027.
Prediction-conditioned BEV feedback for LiDAR detectors: +2.48 Moderate mAP on KITTI (+3.95 pedestrian, +3.44 cyclist), +1.15 L2 APH on Waymo; a LiDAR-only Region-Aware CVAE lowers ECE by 12.7%.

[4] **SmokeGraph-Bench: A Benchmark for Smoking Behaviour Detection with Geometric Relational Baselines**
Md. Irtiza Hossain, Sami Sadat, Junaid Ahmed Sifat, Md. Waseq Alauddin Alvi, Md. Khalilur Rhaman
Under review, Engineering Applications of Artificial Intelligence (EAAI).
A 14,888-image, 167,754-person CCTV benchmark with verified annotator agreement ($\kappa = 0.87$), and RelSmoke, a polar-coordinate graph model (+37% composite score, 157 FPS). [Preprint]

[5] **CLEAR-MoE: Shared-Basis Expert Extraction from Frozen Vision Transformers via Calibration-Driven Layer Selection**
Md. Irtiza Hossain, Humaira Ayesha, Junaid Ahmed Sifat
arXiv:2606.28516, 2026; under review, ICCIT 2026.
Post-training conversion of frozen ViTs into sparse Mixture-of-Experts, retaining 99.9% of dense accuracy across five backbones. [arXiv] [Code]

[6] **Uncertainty-Aware and Efficient 3D Object Detection for Autonomous Driving: A Comprehensive Survey**
Sami Sadat, **Md. Irtiza Hossain**, Md. Golam Rabiul Alam
Manuscript in preparation.
Five-axis taxonomy and quantitative meta-analysis of 127 driving-oriented 3D detection methods.

Selected Projects

Explainable AI for Brain Cancer Classification from MRI Scans [Code]
Applied Grad-CAM and SHAP to CNN classifiers to localise tumour regions and make predictions easier to review.

Five-Class Image Classification with CNNs [Code]
Compared logistic regression, a neural network, AlexNet, ResNet-34, Inception, and a custom CNN on 2,400 imbalanced images, using resampling and augmentation.

Test Scores

TOEFL iBT: 100 / 120

Skills

Deep Learning: PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, Detectron2, MMDetection, Hugging Face
Programming: Python, C, C#, JavaScript, PHP
Tools: Git, Docker, Linux/Bash, LaTeX, Jupyter, Weights & Biases
Web and Backend: React, Node.js, MongoDB, MySQL, .NET

References

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